

TEAM ID – H26SCT01



Neurocrowd AI – Based Real-Time Crowd Intelligence and risk Prediction System

TEAM NAME – AegisVision

TEAM LEAD – SYED ALI FATHIMA N

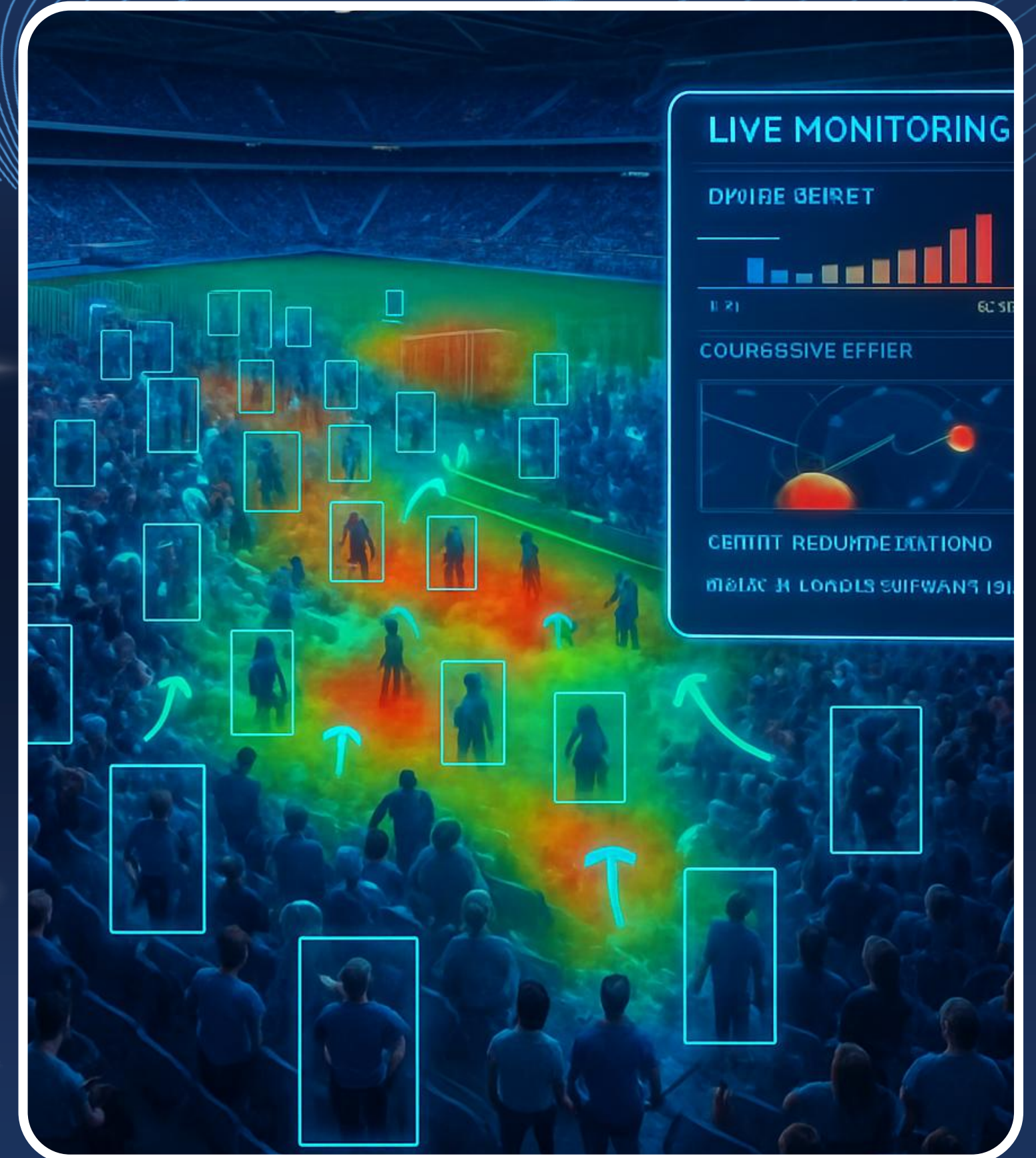
TEAM MEMBER 1 – SOWMIYA N

TEAM MEMBER 2 – MUBEENA BANU M



Problem Statement

Large public venues such as railway stations, festivals, stadiums, and transport hubs face the risk of overcrowding, where traditional CCTV systems only record incidents and rely on manual monitoring. This delays emergency response and increases the possibility of crowd-related accidents. NeuroCrowd addresses this challenge by using AI-powered real-time video analytics to detect people, estimate crowd density, analyze crowd movement, identify congestion zones, predict crowd risk levels, and provide actionable recommendations through a live monitoring dashboard. The objective is to enable proactive decision-making, improve public safety, and support authorities in preventing crowd-related emergencies using existing surveillance infrastructure.



System design



Core Algorithms Used



- ◆ YOLOv8 (Object Detection) Detects and counts people in real-time from CCTV/video streams.
- ◆ Zone-Based Crowd Analysis Computes zone-wise people count and crowd density.
- ◆ Farneback Optical Flow Analyzes crowd movement, motion direction, and turbulence to detect abnormal behavior.
- ◆ Random Forest Classifier Predicts crowd risk levels (Safe, Warning, Critical) using density and motion features.
- ◆ Rule-Based Recommendation Engine Generates real-time alerts and crowd management recommendations based on predicted risk.



Technology Stack



Programming & Development

Python 3.10 was used as the core programming language, with VS Code as the development environment and GitHub for version control and team collaboration.

AI & Machine Learning

YOLOv8 performs real-time person detection, while a Random Forest Classifier predicts crowd risk levels based on density and movement features.

Computer Vision & Analytics

OpenCV handles video processing, and Farneback Optical Flow analyzes crowd movement, direction, and turbulence for abnormal behavior detection.

Dashboard & Data Management

A Streamlit Dashboard provides live crowd monitoring, displaying people count, crowd density, risk level, and real-time recommendations, with YAML and Joblib used for configuration and model management.

Results



KEY OUTCOMES

- Real-Time Person Detection & Tracking
- Crowd Density & Movement Analysis
- Automated Risk-Level Assessment
- Live Monitoring for Early Warning

Existing Solutions and Identified Gaps

Existing Crowd Monitoring Approaches

Most existing crowd surveillance systems rely on conventional CCTV monitoring and basic people detection. These solutions primarily record incidents and depend on continuous human observation, making them reactive rather than proactive. While they provide live video feeds and basic counting capabilities, they offer limited intelligence for anticipating crowd-related emergencies.

Key Gaps

- Reactive monitoring with limited predictive capability.
- Manual supervision required for continuous observation.
- No integrated analysis of crowd density and movement.
- Limited support for early risk identification.
- Minimal decision support for emergency response teams.



Our Innovation – NeuroCrowd

AI-Powered Predictive Crowd Intelligence

NeuroCrowd transforms traditional surveillance into an intelligent decision-support system by combining computer vision, crowd analytics, and AI-based risk prediction. Instead of only detecting people, the system continuously analyzes crowd density, movement patterns, and zone-wise congestion to estimate risk levels and provide actionable recommendations in real time.

Key Innovations

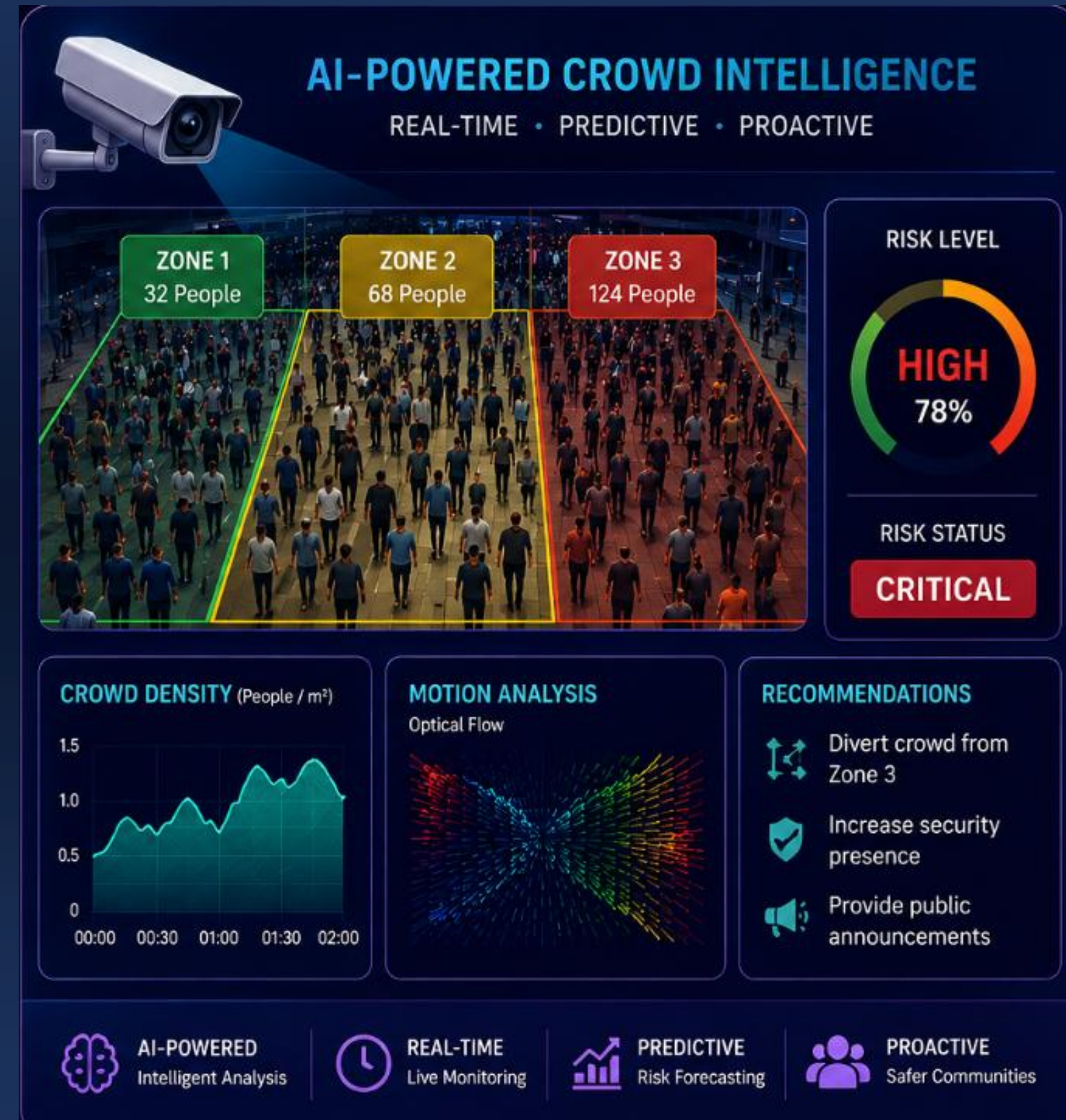
- Real-time AI-powered crowd monitoring.
- Zone-wise crowd density estimation.
- Motion and turbulence analysis using Optical Flow.
- AI-based risk prediction (Safe, Warning, Critical).
- Instant recommendations for proactive crowd management.
- Works with existing CCTV infrastructure, enabling cost-effective deployment.

Summary & Future Work

NeuroCrowd is an AI-powered crowd intelligence system that enhances public safety through real-time people detection, zone-wise crowd density analysis, motion pattern analysis, and AI-based risk prediction. By transforming conventional CCTV surveillance into a proactive decision-support system, it enables authorities to identify potential crowd risks early and take preventive action before situations escalate.

Future Work

- Improve detection accuracy using advanced YOLO models and multi-object tracking.
- Train the risk prediction model with large-scale real-world crowd datasets.
- Integrate automatic alert systems (SMS, mobile notifications, and control room integration).
- Support multi-camera monitoring with a centralized dashboard.
- Deploy on edge AI devices for faster, real-time performance in smart city environments.





THANK YOU